

Advancing Neutral Atom Trapping Through the Development of Digital Laser Control Systems

Undergraduate Thesis
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1 Abstract

Precise control of laser wavelength is an essential step for neutral atom trapping, which can be achieved with a digital dichromic atomic vapor laser lock controller, developed to calibrate 780-nm wavelength laser light for use in trapping rubidium-87 atoms in a magneto-optical trap. By collecting the response from a sample of rubidium using photodiodes connected to an Arduino Due communicating with a program developed in Python, this controller is able to search for and track target responses and create a closed-loop control for the laser during setup and operation. With a priority on low cost and a high level of autonomy, this system allows for control of laser wavelength to a precision of 0.01 nm while reducing overhead for the experiment's operation.

2 Introduction

Quantum computing is an exciting new area of study capable of significantly changing the way that computing is done through increasing the speed and efficiency of computing, by providing an exponential speedup to some calculations and aiding the creation of physical simulations previously believed to be infeasible to create.

Neutral atom optical traps are prevailing as an effective way to develop quantum computers. Using the electronic and nuclear spin of these atoms, the atoms can be trapped, prepared, and read using lasers tuned to a precisely to a desired wavelength.

These neutral atom quantum computers allow for the development of quantum gates in a practical way, aided by the longer decoherence times due to the neutral atom's properties [7].

This project is a step forward in creating a system to allow for the preparation of neutral atoms for use in quantum gates. Over the course of one year, with the help and advising of Dr. Jay Erker, the work on this project has resulted in an autonomous controller for the Dichromic-Atomic Vapor Laser Lock(DAVLL) system used for preparing rubidium atoms in the Gillen Cold Atom Project.

2.1 Gillen Cold Neutral Atom Quantum Computer

The previous work in this project involved developing a neutral atom trap for quantum computing using atoms of rubidium-87, controlled using a system of 780nm-wavelength laser light. The cold atoms are contained in bright-spot traps, where the precision of the laser light tuning is a necessity to trap and manipulate atoms properly[1].

A significant portion of the experiment is dedicated to calibration and control of the trapping laser. These open diode-lasers are controlled through the current, operation temperature, and angle of diffraction in the laser cavity.

The laser light is generated using a 780-nanometer laser diode controlled by a Thorlabs ITC510 laser controller, to control the operating current and temperature of the laser diode. Operating current controls the total power of the laser, a requirement for both the light intensity and the control of the laser mode. The temperature is controlled by a thermistor, quantified by the thermistor's resistance, which keeps the laser operating at a desired frequency.

This controller is connected to an open diode laser, where the laser light is dependent on the light diffracted into the laser chamber. This diffraction angle is controlled with a piezoelectric device connected to a diffraction grating, allowing for physical movement of the grating with an applied voltage.

2.1.1 Laser Grating Feedback

The laser is calibrated using the response of a sample of a rubidium vapor to laser light at 780nm. As shown in figure 1, a portion of the beam is split and polarized, which goes through the sample of rubidium in a magnetic field. The response enters a beamsplitter that feeds the two polarized responses, σ^+ and σ^- , into photodiodes.

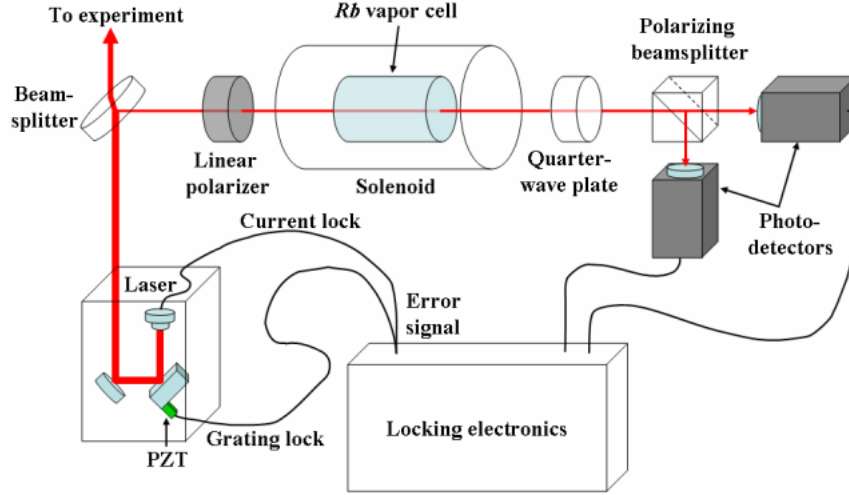


Figure 1: Setup of DAVLL on Laser Table
[1]

Once the laser is tuned[3], Figure 2 shows the response of the rubidium when the laser is at the correct frequency. A challenge with the experiment is that during operation, the wavelength has a tendency to drift. Diagram (c) in figure 2 shows that the combination of these signals results in a desired "zero-point" that can be used to correct for drift by centering the zero-point. This is elaborated on in section 4.2.3.2 of Dr. Gillen's thesis[1].

2.2 Development of the Ramp Controller

The focus of the project is to create tools that allow for a more efficient and streamlined experiment startup and control, allowing for experimenters to put focus on other parts of the project during runtime.

The beginning of the process for developing an autonomous system involved developing a digital ramp controller.

While the laser is operating, a voltage is applied to the piezoelectric which increases linearly in a periodic fashion, creating the voltage "ramp" to maneuver the grating discussed in section 2.1. While the diode mounting frame allows for coarse adjustments, and the current and temperature control of the laser allows for fine adjustments, the piezoelectric allows for very fine control of the laser wavelength, down to a hundredth of a nanometer, giving the precision necessary for the experiment.

In previous work on this project, the ramp voltage controller was digitized, increasing the sweep range from 12 Volts to 70 Volts, sweep range, as well as allowing digital control of parameters such as the ramp period and necessary ramp attenuation.

An arduino due connected to a digital-to-analog converter(DAC) runs a code loop that increases the voltage over a set period of time. This signal is then when amplified to the desired voltage, creating the ramp signal.

This design was then printed onto a custom circuit board connected to an arduino due, as shown in figure 3. This was a major step forward in the project, as it allowed for faster setup process compared to previous designs.

A constant ramp from zero voltage to maximum, however, was not enough to lock the laser. Greater control over the laser, as well as the ability to respond to feedback from the rubidium vapor, was necessary.

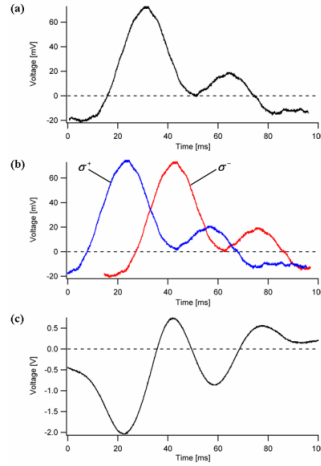


Figure 2: Rubidium response to 780nm laser light. (a) shows the response of outside of a magnetic field, (b) shows the polarized response in a magnetic field, and (c) shows the result of $\sigma^+ - \sigma^-$ from (b).
[1]

3 Project Goals

The intention of this project was to introduce the input of feedback into the system, allowing for closed loop control of the laser, and giving the experiment the ability calibrate and correct for drift.

The previous method of calibrating the laser involved reading the response of the rubidium into an analog controller, which used the "zero point" of the response in figure 2 (c) and its position as a control for the voltage of the piezoelectric.

While this was effective, the project has been significantly aided through the automation and digitization of this control system, as it allows for control through an external program, and the abstraction of the experiment control into a single interface, when combined with the laser controls and visualization of the rubidium response.

The focus of this project has been to create a digital system to manage setup and drift response, of the system. By recording the rubidium fluorescence in a digital format and processing it in an external program, combined with the digital ramp and the laser controllers, the potential of the experiment for autonomy and capability of future projects is opened up significantly, reducing the overhead required to run the experiment, and allowing for errors to be corrected before they become noticeable.

This project used the introduced functions that require very little input from the user to set up and control the experiment, including:

Data Collection Sweep through a range of current and temperature settings and record the response at each, so that they can be used for calibration and development of future systems.

Drift Correction A major issue previously discussed is the laser's tendency to drift. Compensate for this drift by tracking the center point of the rubidium fluorescence, and changing the ramp position or the laser settings as necessary.

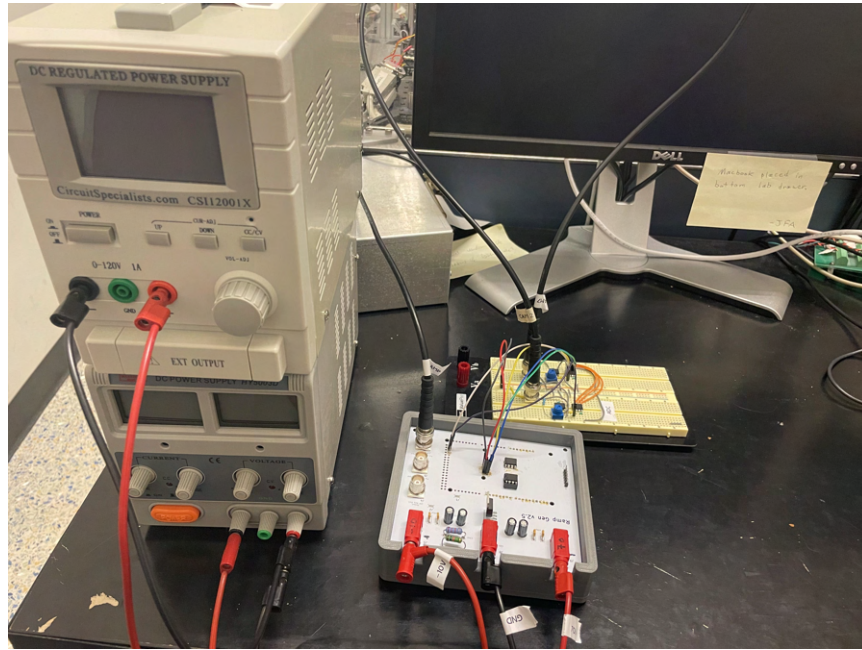


Figure 3: Setup of Ramp Board with Power Supplies.
[3]

Experiment "Cold Start" Prepare the laser for the experiment by locating laser settings that result in a rubidium response. Iterate through a wide range of settings until a response is found, and then refine settings until the response is strongest.

3.1 Requirements

In order to make these programs feasible, the inclusion of several new features needed to be developed:

Read Photodiode Feedback The photodiodes mentioned in section 2.1.1 output an analog response that needs to be captured digitally. A system needed to be developed to record this response simultaneously with the ramp, synchronized with the ramp position.

Control and Monitor Laser Autonomous control of the laser's settings was a critical requirement for both error correction and locating good signals. Additionally, being able to control the laser settings automatically offers more precision when setting the laser.

Filter and Locate Desired Signal Once the signal is stored in the program, being able to determine if the signal matches the response of rubidium fluorescence is critical. Through a combination of smoothing, normalization, and other digital signal processing, locate and track this response.

While laser controller solutions for neutral atom quantum computing currently exist, their high costs result in a high barrier to entry for experimentation. This project was designed with the following priorities:

Simplicity The project was designed and built with components that are low cost and common. Not only does this make each component easy to replace, but it allows for other groups to build this as well, lowering the barrier of entry for quantum computing.

Accuracy Despite the simple components, accuracy is still critical to the projects success. High fidelity of the responses as well as a large number of samples are both key to accurately modeling the response and controlling the project in an effective way.

Upgradability The abstraction of components of the project allows both the hardware and software to be upgraded later. This is critical for allowing the project to grow easily, be repaired quickly, and allow for new functionality with requiring an entirely new experimental setup.

Compatibility The project needed to be built in an open-source using common communication standards, allowing for the ability to interact with other programs and components. In addition, it needed to be able to support working with the custom components of the experiment.

4 Photodiode Reader

The first component to be developed was the photodiode reader, a circuit used to capture input from the response detectors discussed in section 2.1.1.

The first design relied a mutli-step analog signal processing circuit where the response was first amplified, then attenuated using a digitally-controlled potentiometer. Once both signals were tuned to a desired amplitude, σ^- was inverted, and the result was added to σ^+ . This final analog signal was then converted into a digital signal and stored in the arduino.

This initial design involved passing each signal through up to 4 separate components after the photodiode reader. Each separate component added complexity that reduced the quality of the signal and increased the chances of a faulty components.

To simplify, the final design directly reads the signal from each detector using an MCP3202 analog-to-digital converter (ADC) controlled via Serial Peripheral Interface(SPI) by the Arduino. This was decided to be the best way to collect input as it maintains the original signal while still having a high enough resolution to preserve it. The output fluorescence range is approximately $100mV$, and the maximum resolution of the 12-bit ADC with a 5V supply is $1.22mV$, giving enough resolution.[4]

To collect data, the arduino polls the voltage on the ADCs for each voltage increment of the ramp. This creates a complete picture of the laser response in the arduino's memory.

The arduino's connection to a PC allows for continuous reading of the recorded data. The collection of this data, in combination with the ramp controller, enables capturing of the laser response and storing the data in a log file to be analyzed through the methods discussed in section 7.

5 Combined Arduinos

With the system for collecting response data developed, programs were be created that interact with both the digital ramp and the photodiode reader. While this system was effective, it required the synchronization of the ramp with the reader to preserve the context of the recorded signal.

The first solution developed was an interrupt-based loop, where the ramp controller signalled the start of a ramp, enabling the program to poll the feedback reader until the ramp controller signaled the end of the ramp. This loop ran continuously to store data.

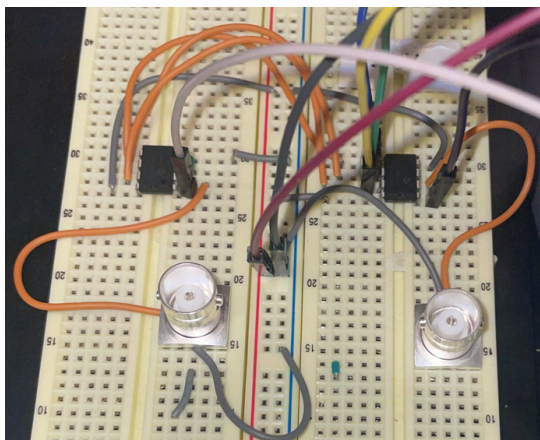


Figure 4: Photodiode reader implemented on a breadboard.

It was eventually decided that it would be advantageous to have the ramp control and feedback reader managed by a single arduino. In addition to reducing the complexity of the devices, it allowed for the synchronization of data collection to be managed by one arduino, reducing time delay and increasing response fidelity, as well as reducing communication overhead.

5.1 Implementation

This was managed by connecting existing pins on the ramp board to the photodiode reader circuit, and soldering new pins as necessary. Chip select pins and data-out pins were added to allow full functionality.

Once connected, issues with the digital signal occurred. It was determined that the analog signal from the photodiodes was interfering with the connection to the computer, causing issues.

To counteract this, the code was refactored in a way where reading the collected data does not happen when a response is being received from the photodiode. This can be improved by impedance matching the components and verifying no wiring or grounding issues exist by printing a circuit board.

6 Code

The python library, developed to allow for data collection and laser control, was designed in an object-oriented fashion where each device was controlled by its own class, and the subsequent classes were then controlled by a command line interface.

When using two separate arduinos to record data, the ramp and response signals were awaited by the program, which, upon receiving a ramp start(valley) signal, began polling the photodiode reader for data, recording the levels of the response constantly. Once the ramp end(peak) signal was received, the system allows for complete feedback to be sent to a log for the specific run.

This was later revised to give more read control to the connection manager and greater autonomy to the arduino by refactoring the format in which data is sent. Instead of the response data being sent immediately after each ramp generation, the controller checks for a request for data at the end of each ramp. If a request has been sent, the controller sends a response in the format shown in table 1.

Length n of arrays	PD Response Array 1	PD Response Array 2
16 bits	$n \times 16$ bits	$n \times 16$ bits

Table 1: Format of response data sent to program.

The loop for collecting data runs constantly while the ramp is connected on a separate thread, storing the formatted response in a global variable. The user has the option to record all data in the log file, or collect the data for use in other methods with a blocking function that returns the most recent response.

6.1 Automatic Data Collection

The automation of the ramp control, as well as the collection of the rubidium response and storing it into a log opened up the opportunity for data from various laser settings to be collected.

In order to be able to locate strong target settings for the laser, the response of rubidium at various settings was required to understand what desired settings for the laser controller were, and to develop a system to score and locate desired settings autonomously.

Previous work on this project[3] enabled the ability to control the external laser power supply and controller[6] using python code. As a result, changing the current and resistance settings for the diode laser was included in the autonomous control loop.

A library of functions was created in the LaserController class. With the usage of a General Purpose Interface Bus (GPIB) connection to the controller, the program has the ability to read and set the current and resistance control in conjunction with the ramp.

6.2 Live Data Graphing

The response data, updated and located in a global variable, enabled the ability to graph the data as it was updated. Using a matplotlib interface in an external thread, the user can view the response of the detectors in real time, allowing for a view of the response at the time and removing the need for the inclusion of an oscilloscope. The graph also gives a visual check to verify that the incoming data is expected for the given laser settings.

6.3 Command Line Interface

As the codebase began to increase in complexity, a system for managing the various processes was required. A command line interface was developed, capable of completing all necessary functions to manage inputs and feedback, interact with the ramp and laser controller, visualize the results from experiment, and allow the user to run autonomous or controlled functions.

The program is developed using the cmd2 library[2], which creates a baseline for a command line interface. This interface is significant as it allows for interaction with the program that does not require coding to run unique sequences. Each control is available as a command that the user can enter, with specific inputs that can be verified. The program is also capable of handling errors in way that does not crash the program and can keep the controller running continuously. In addition, adding new commands is a simple process that allows the scope and abilities of the project to be extended easily.

The interface uses a python thread library to have the ability to run multiple tasks at once. This is crucial to the successful operation of the experiment. As the data from a ramp is loaded from the Arduino, the program can

monitor the laser controls, interactively graph the ramp output, and run autonomous control functions, all while the user enters commands.

This results in a sequential mix of autonomous and user-controlled operations, all while the program manages the laser properly. This simplifies the system enough that much less time needs to be spent on setup of the experiment.

7 Data Collection

The ability to control the laser while collecting the rubidium response was key to begin develop tools to assist in the experiment. The first autonomous loop, to collect laser response data, was developed to achieve the goal listed in section 3.

This program sets the laser to a specific target current and resistance, and records the response of the system when the ramp iterates through its entire range.

The control loop involves first setting the resistance to a target, and then setting the current. Once the response has been recorded, it moves through all specified current values. Upon completion, the program then moves to the next resistance level, and repeats until all possible combinations (within a specified granularity) have been recorded. The programmer can specify the starting, ending, and value to increment by for the current and resistance.

The resultant data is significant not only as it allows for an understanding of how each of these controls changes the laser's properties, but it also allows for fast access of the previous data and the ability to see the response from previous settings.

In order to see the data, a program was developed to load the data, display it on a graph, and add controls to quickly view and compare laser set points, and to locate specific responses from the rubidium. Figure 5 shows the graphs displayed and the ability to control them.

This allows for a way to quickly visualize and understand the response of the rubidium throughout a range of settings as broadly defined and as finely tuned as the user specifies. With this, finding good operating points for the laser or noticing patterns within the laser response becomes much easier, especially as the response data from even short runs can be megabytes in size.

8 Filtering and Scoring

The collection of data at various laser current and temperature settings mentioned in section 7 not only allows for visualization, but for the ability to analyze the data and develop more functionality.

As stated in section 3, a primary focus of this project is digital control of drift. While user-controlled laser calibration is an effective way to setup the laser, and the digital controls facilitate this process, the ultimate goal of the project is to have the ability to correct for error in the signal without the need of human intervention.

In order to achieve this, the data needs to be filtered and converted into an error signal to be used as a control.

8.1 Signal Filtering

Due to imperfections in the input signal, code-based signal processing was necessary to use the signal as a control. Using filtering libraries in Python was more feasible than implementing hardware-based filters as they can be easily swapped, are more reliable, and allow for preservation of the original signal for further manipulation.

The individual signals σ^+ and σ^- vary in both offset and amplitude. In order to compare these signals, they are z-normalized.

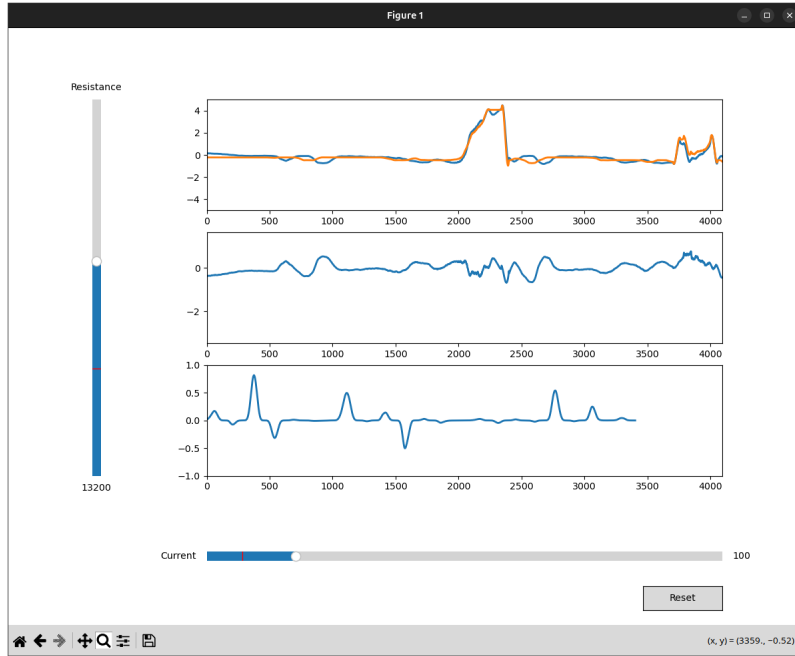


Figure 5: Example output of the feedback viewer program. The top graph shows the smoothed output, the middle graph shows the combination of the two signals, and the bottom graph shows the signal's "score" by the windowing program. The bottom and right side are the sliders for selecting the resistance and current.

The digitization of the signal using the photodiode reader results in some loss of signal quality, and a roughness in the signal that, while preserving the general shape of the signal, has slight differences when compared to the feedback of rubidium fluorescing, shown in figure 6.

Thus, the signal is processed using a Savitzky-Golay filter[5], using 3rd-order polynomials and a window size of 60 samples. The resultant signal is an output that more closely resembles the target signal, reducing noise and loss of signal fidelity.

Once these signals have been normalized and filtered, the difference between the two can be easily found.

8.2 Windowing and Scoring

In order to achieve the goal listed in section 3, the program needs to be able to have a quantifiable score for the resultant output, and know where the rubidium response is located along the ramp.

The program uses a sliding window to compare the ramp output to a target response, giving a score between -1 and 1 on the closeness to the template of the result in that window, where 0 is the least close, 1 is an exact match, and -1 is an exact inverse.

To improve the system, the output is cubed in order to reduce the magnitude of mid-range scores, which typically occur inside mode humps. This also preserves the sign of the score, showing when the signal is inverted.

The output of this score is shown in Figure 7. Note the similarity between the top graph and the combined

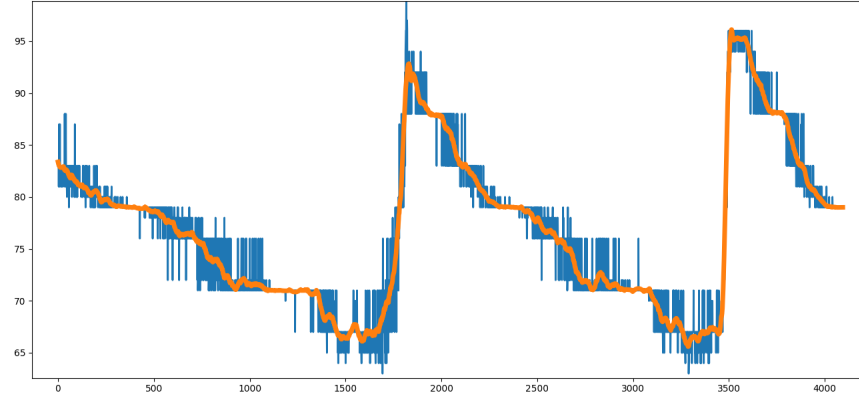


Figure 6: The input from a photodiode reader (blue) and the smoothed output.

signal shown in Figure 2c. The bottom graph then shows a peak (close to 1) at the start of the signal, which can be used as a reference point for the response.

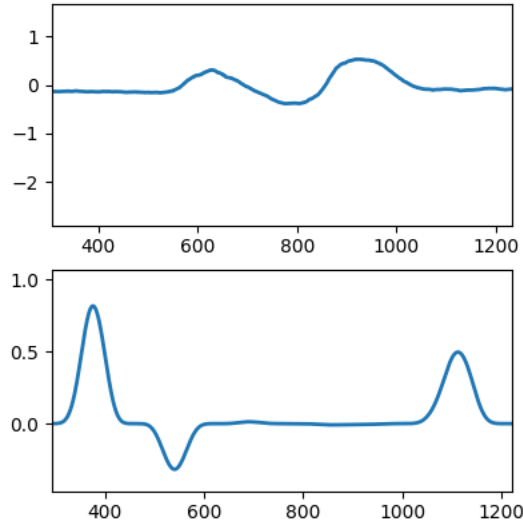


Figure 7: Demonstration of the combined response signal at a target point (top) and its subsequent score (bottom).

This targeting system is crucial to the success of the project. With a single value as input, the system for locating strong rubidium responses across a wide range of laser settings becomes a system that can be completely managed by the program. This opens up the possibility for the system to find these high-response settings, then

improve the settings through more fine-grained control of the laser. As a result, the system can calibrate and set the laser with minimal input necessary from the user.

9 Wavelength Widths and Progress on Measurements

Previous versions of the ramp had a 12V range[1], which was only enough to see one response from the rubidium. However, the current voltage range of 70V allows for a significantly larger range of responses, and the ability to see multiple rubidium responses on a single period of the ramp.

The digital controller output has a voltage step size of 0.024%, allowing for control of the wavelength of the laser with precision of hundredths of nanometers.

The result of these two factors — a wide-ranging ramp with very fine precision — gives the best of both worlds, and is significant step forward in this project. The ability to see multiple responses from the rubidium and manually select the most usable without needing to change the laser settings and potentially losing the response gives much greater control over further steps in the experiment.

10 Conclusion

This project was a major step forward in the automation of the cold atom trap experiment. Integrating the digital ramp controller, which already provided greater functionality to the project, into a control loop that tracks the response of the rubidium creates a system that facilitates the experiment while keeping the barrier of entry low.

The ability to collect and store the response digitally, while tracking the laser status and synchronizing with the ramp position, is also a major benefit for the project. Being able to see how previous settings affected the response is highly beneficial for calibrating the laser and analyzing the response using quantitative data after a run of the experiment has occurred.

The scoring system using a sample window is a useful way to quantify a response into single value that can be then used as a control. Additionally, the target sample to compare the signal against can easily be swapped inside the program, allowing for even more uses for the program.

This is an example of the modularity that was prioritized in this project — which makes upgrading the project in the future possible without limiting current functionality. The command line interface that the control program is built around allows for the addition of new commands with just the development of a new function. In addition, each component acts in an independent manner that allows for replacement of components when deemed necessary.

While this project is a step forward, more work can still be done to improve the quality of the project, through both the autonomous programs and the hardware for the controller.

Now that the controls are digitized, the ability to start the experiment and calibrate the laser through a broad search of settings until a strong response is located can easily be implemented through the abstraction offered in the code. Additionally, the scoring system, when calibrated to control the laser settings, will allow for the correction of drift.

For the hardware, improving the quality of the photodiode reader, and potentially amplifying the signal before digitizing it will allow for better control. This will also be improved when the system is printed onto a PCB verifying good connections.

Continued work on this project as well as its replication for use in other atom trap experiments will further facilitate the work in field of atom trapping, and as a result, quantum computing. In addition, this device reduces the barrier of entry for other experiments, enabling more work in this field.

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